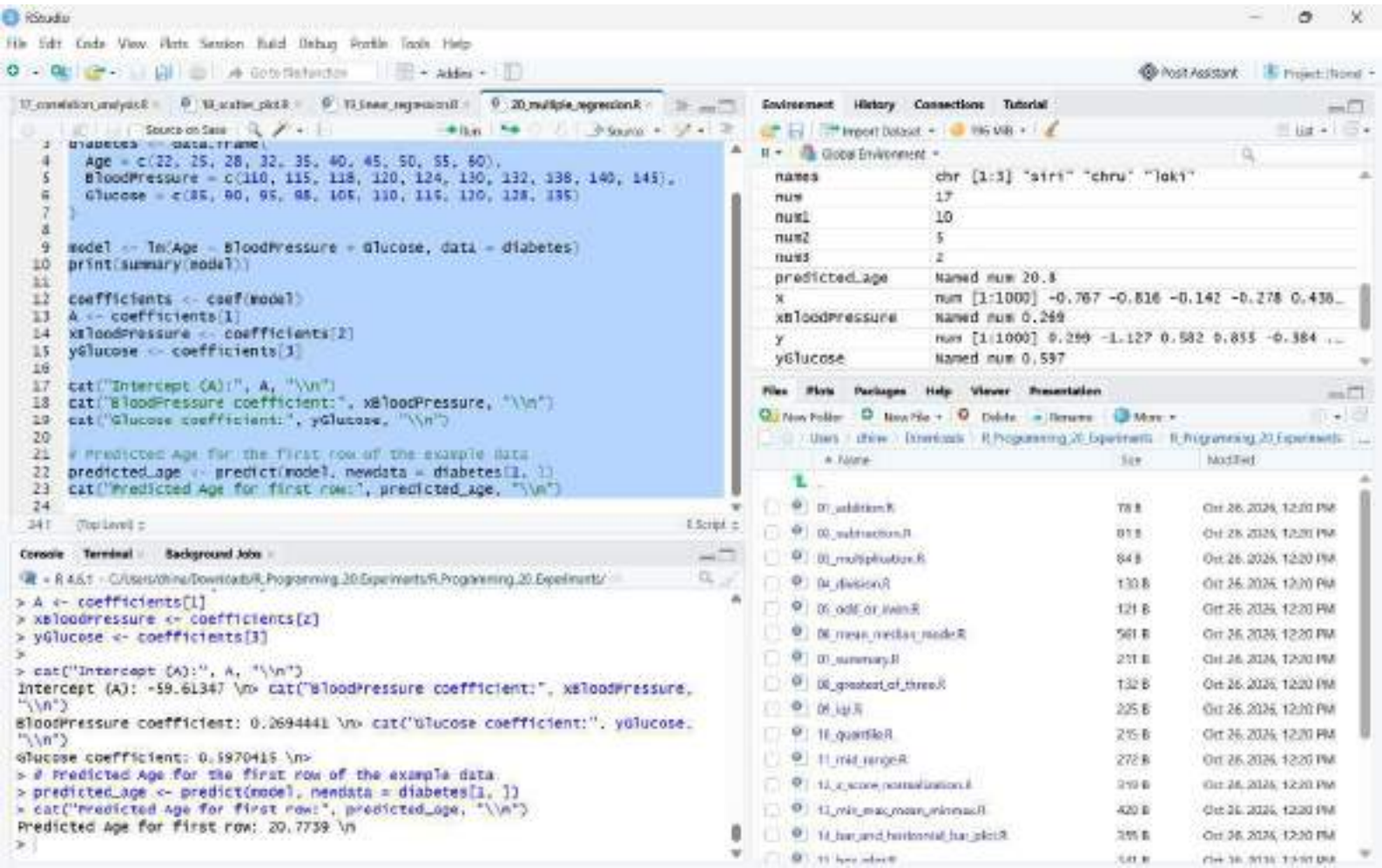

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
17_correlation_analysis.R
18_scatter_plot.R
19_linear_regression.R
20_multiple_regression.R
Source on Save
Run
Source
diabetes <- data.frame(
Age = c(22, 25, 28, 32, 35, 40, 45, 50, 55, 60),
BloodPressure = c(110, 115, 118, 120, 124, 130, 132, 138, 140, 145),
Glucose = c(85, 90, 95, 98, 105, 110, 115, 120, 128, 135)
)
model <- lm(Age ~ BloodPressure + Glucose, data = diabetes)
print(summary(model))
coefficients <- coef(model)
A <- coefficients[1]
xBloodPressure <- coefficients[2]
yGlucose <- coefficients[3]
cat("Intercept (A):", A, "\\n")
cat("BloodPressure coefficient:", xBloodPressure, "\\n")
cat("Glucose coefficient:", yGlucose, "\\n")
Predicted Age for the first row of the example data
predicted_age <- predict(model, newdata = diabetes[1,])
cat("Predicted Age for first row:", predicted_age, "\\n")
(Top Level)
R Script
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Terminal
Background Jobs
R 4.5.1 · C:/Users/dhina/Downloads/R_Programming_20 Experiments/R_Programming_20 Experiments/
> A <- coefficients[1]
> xBloodPressure <- coefficients[2]
> yGlucose <- coefficients[3]
>
> cat("Intercept (A):", A, "\\n")
Intercept (A): -59.61347 \n> cat("BloodPressure coefficient:", xBloodPressure, "\\n")
BloodPressure coefficient: 0.2694441 \n> cat("Glucose coefficient:", yGlucose, "\\n")
Glucose coefficient: 0.5970415 \n>
> # Predicted Age for the first row of the example data
> predicted_age <- predict(model, newdata = diabetes[1,])
> cat("Predicted Age for first row:", predicted_age, "\\n")
Predicted Age for first row: 20.7739 \n
>
Environment History Connections Tutorial
Import Dataset
Global Environment
names chr [1:3] "siri" "chru" "loki"
num 17
num1 10
num2 5
num3 2
predicted_age Named num 20.8
x num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ...
xBloodPressure Named num 0.269
y num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ...
yGlucose Named num 0.597
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Name
Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 581 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_sqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 420 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 395 B Oct 26, 2026, 12:20 PM

RStudio

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```
# Experiment 19: Linear Regression
# Example data is used instead of an external diabetes dataset
diabetes <- data.frame(
  Age = c(22, 25, 28, 32, 35, 40, 45, 50, 55, 60),
  BloodPressure = c(110, 115, 118, 120, 124, 130, 132, 138, 140, 145)
)

model <- lm(BloodPressure ~ Age, data = diabetes)
print(summary(model))

plot(diabetes$Age, diabetes$BloodPressure,
     col = "darkgreen", pch = 19,
     main = "Linear Regression Analysis",
     xlab = "Age", ylab = "Blood Pressure")
abline(model, col = "red", lwd = 2)
```

```
Age          0.88922    0.03298   26.96 3.86e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

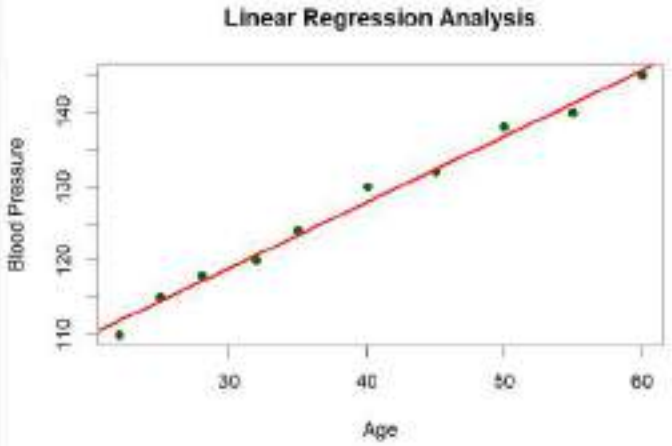
Residual standard error: 1.388 on 8 degrees of freedom
Multiple R-squared:  0.9891,    Adjusted R-squared:  0.9878
F-statistic: 726.8 on 1 and 8 DF,  p-value: 3.859e-09

>
> plot(diabetes$Age, diabetes$BloodPressure,
+      col = "darkgreen", pch = 19,
+      main = "Linear Regression Analysis",
+      xlab = "Age", ylab = "Blood Pressure")
> abline(model, col = "red", lwd = 2)
>
```

Environment History Connections Tutorial

model	List of 12
Values	
a	num [1:5] 55 67 89 80 90
age	num [1:3] 23 24 25
age_group	chr [1:12] "Young" "Young" "Middle" "Middle" "...
age_mean	30.2
age_sd	7.59605160593318
frequency_table	'table' int [1:3, 1:3] 2 2 0 0 1 3 2 1 1
greatest	42
insulin_group	chr [1:12] "Low" "Low" "Normal" "High" "High" ...

Files Plots Packages Help Viewer Presentation



RStudio

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```
# Experiment 18: Scatter Plot
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)

par(mfrow = c(1, 2))
smoothScatter(y - x, main = "Smoothed Density of y - x",
              xlab = "Index", ylab = "y - x")
smoothScatter(x, y, main = "Scatter Plot", xlab = "x", ylab = "y")
par(mfrow = c(1, 1))
```

Environment History Connections Tutorial

mid_range	24
min_max	num [1:6] 0 0.211 0.474 0.737 1
minimum	21
names	chr [1:3] "siri" "chru" "loki"
num	17
numL	10
numZ	5
num3	2
x	num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436…
y	num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 …

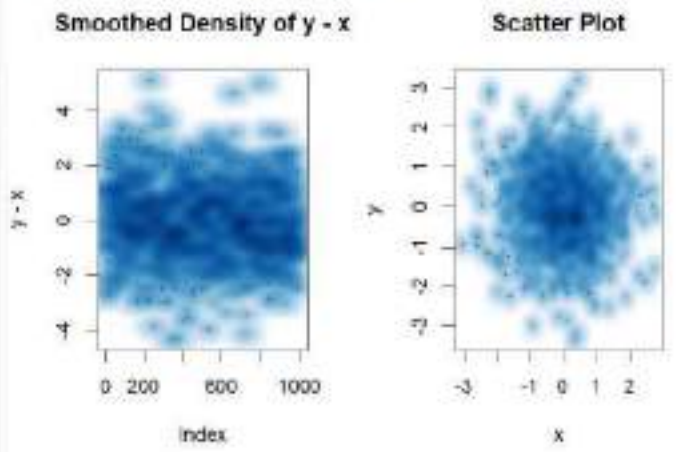
Files Plots Packages Help Viewer Presentation

Console Terminal Background Jobs

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```
Warning message:
In chisq.test(frequency_table) : Chi-squared approximation may be incorrect

> # Experiment 18: Scatter Plot
> set.seed(9)
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> par(mfrow = c(1, 2))
> smoothScatter(y - x, main = "Smoothed Density of y - x",
+               xlab = "Index", ylab = "y - x")
> smoothScatter(x, y, main = "Scatter Plot", xlab = "x", ylab = "y")
> par(mfrow = c(1, 1))
>
```



RStudio

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14_bar_and_horizontal_bar_plot.R | 15_box_plot.R | 16_histogram.R | 17_correlation_analysis.R

Source on Save | Run | Source

```
# Experiment 17: Association Analysis using a Chi-square Test
# Example categorical data is used instead of an external Excel file.
age_group <- c("Young", "Young", "Middle", "Middle", "Older",
               "Older", "Young", "Middle", "Older", "Young",
               "Middle", "Older")
insulin_group <- c("Low", "Low", "Normal", "High", "High",
                   "Normal", "Normal", "High", "Low", "Low",
                   "Normal", "High")

frequency_table <- table(age_group, insulin_group)
print(frequency_table)
print(chisq.test(frequency_table))
```

(Top Level) | R Script

Console | Terminal | Background Jobs

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```
+                   "Normal", "High")
>
> frequency_table <- table(age_group, insulin_group)
> print(frequency_table)
          insulin_group
age_group High Low Normal
  Middle    2   0      2
  Older     2   1      1
  Young     0   3      1
> print(chisq.test(frequency_table))

        Pearson's Chi-squared test

data:  frequency_table
X-squared = 6, df = 4, p-value = 0.1991
```

Environment | History | Connections | Tutorial

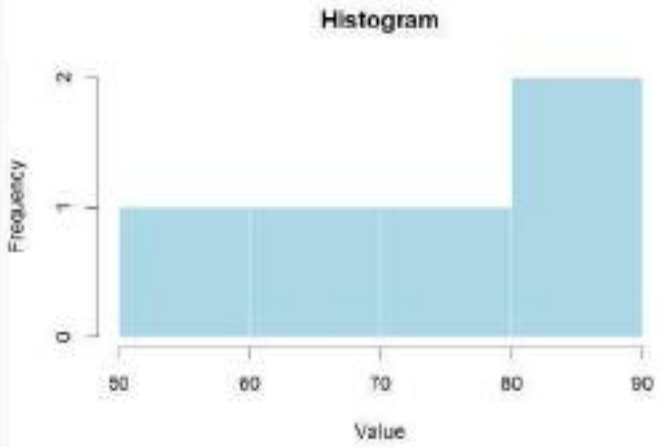
Import Dataset | 180 MiB | List

R | Global Environment

age_group	chr [1:12] "Young" "Young" "Middle" "Middle" "...
age_mean	50.2
age_sd	7.596051605933318
frequency_table	'table' int [1:3, 1:3] 2 2 0 0 1 3 2 1 1
greatest	42
insulin_group	chr [1:12] "Low" "Low" "Normal" "High" "High" ...
marks	num [1:5] 88 78 25
maximum	40
mean_age	50.2
mid_range	24

Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Fit to Pane | Export | Publish



RStudio

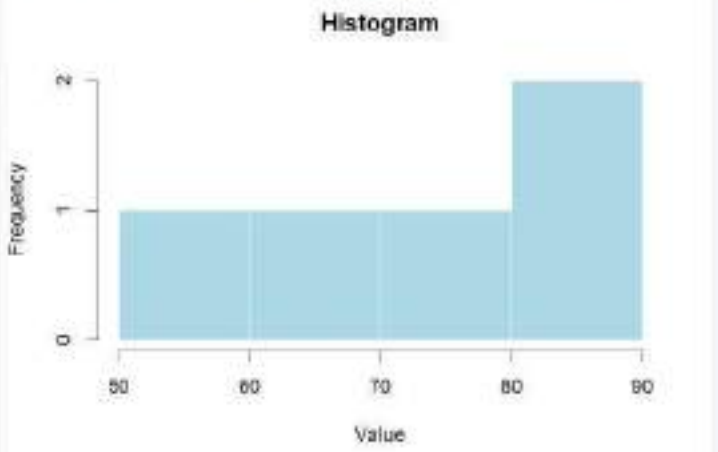
File Edit Code View Plots Session Build Debug Profile Tools Help

```
# Experiment 16: Histogram
a <- c(55, 67, 89, 80, 90)
hist(a, main = "Histogram", xlab = "value", col = "lightblue",
     border = "white")
```

```
> # Experiment 15: Histogram and Box Plot
> names <- c("siri", "chru", "lok1")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> par(mfrow = c(1, 2))
> hist(df$age, main = "Age Histogram", xlab = "Age", col = "lightblue")
> boxplot(df$age, main = "Age Box Plot", ylab = "Age", col = "lightgreen")
> par(mfrow = c(1, 1))
> # Experiment 16: Histogram
> a <- c(55, 67, 89, 80, 90)
> hist(a, main = "Histogram", xlab = "value", col = "lightblue",
+      border = "white")
>
```

Values

a	num [1:5] 55 67 89 80 90
age	num [1:3] 23 24 25
age_mean	90.2
age_sd	7.590051605933118
greatest	42
marks	num [1:3] 88 78 25
maximum	40
mean_age	90.2
mid_range	24

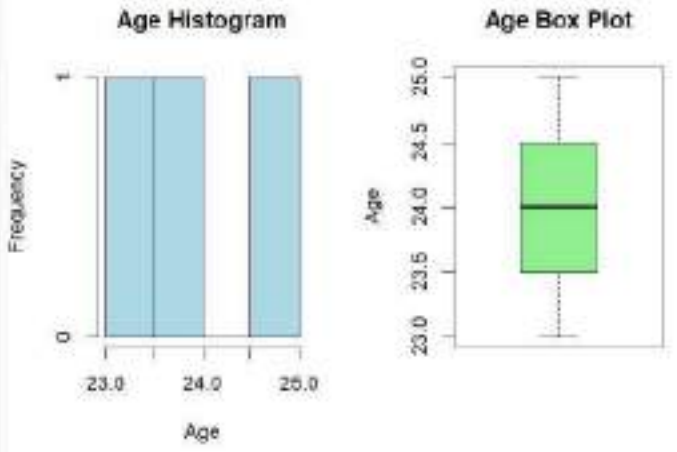


```r
# Experiment 15: Histogram and Box Plot
names <- c("siri", "chru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)

par(mfrow = c(1, 2))
hist(df$age, main = "Age Histogram", xlab = "Age", col = "lightblue")
boxplot(df$age, main = "Age Box Plot", ylab = "Age", col = "lightgreen")
par(mfrow = c(1, 1))
```

```
+         names.arg = paste0("Item ", seq_along(a)))
> barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
+         xlab = "Value", names.arg = paste0("Item ", seq_along(a)))
> par(mfrow = c(1, 1))
> # Experiment 15: Histogram and Box Plot
> names <- c("siri", "chru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> par(mfrow = c(1, 2))
> hist(df$age, main = "Age Histogram", xlab = "Age", col = "lightblue")
> boxplot(df$age, main = "Age Box Plot", ylab = "Age", col = "lightgreen")
> par(mfrow = c(1, 1))
>
```

age_mean	30.2
age_sd	7.696051605933118
greatest	42
marks	num [1:3] 88 78 25
maximum	40
mean_age	30.2
mid_range	24
min_max	num [1:5] 0 0.211 0.474 0.737 1
minimum	21
names	chr [1:3] "siri" "chru" "loki"



RStudio

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```
# Experiment 14: Vertical and Horizontal Bar Plots
a <- c(55, 67, 89, 80, 90)

par(mfrow = c(1, 2))
barplot(a, main = "Bar Plot", xlab = "Item", ylab = "Value",
        names.arg = paste0("Item ", seq_along(a)))
barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
        xlab = "Value", names.arg = paste0("Item ", seq_along(a)))
par(mfrow = c(1, 1))
```

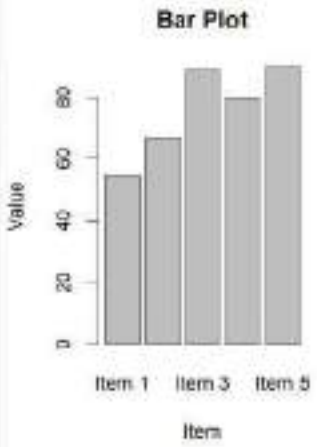
Environment History Connections Tutorial

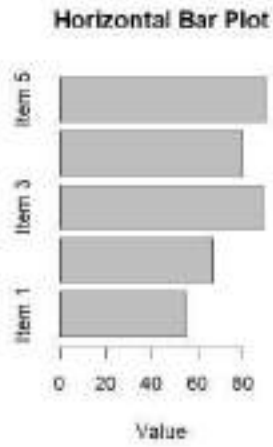
Global Environment

values

a	num [1:5] 55 67 89 80 90
age	num [1:5] 21 25 30 35 40
age_mean	30.2
age_sd	7.590051605933118
greatest	42
marks	num [1:3] 88 78 25
maximum	40
mean_age	30.2
mid_range	24

Files Plots Packages Help Viewer Presentation

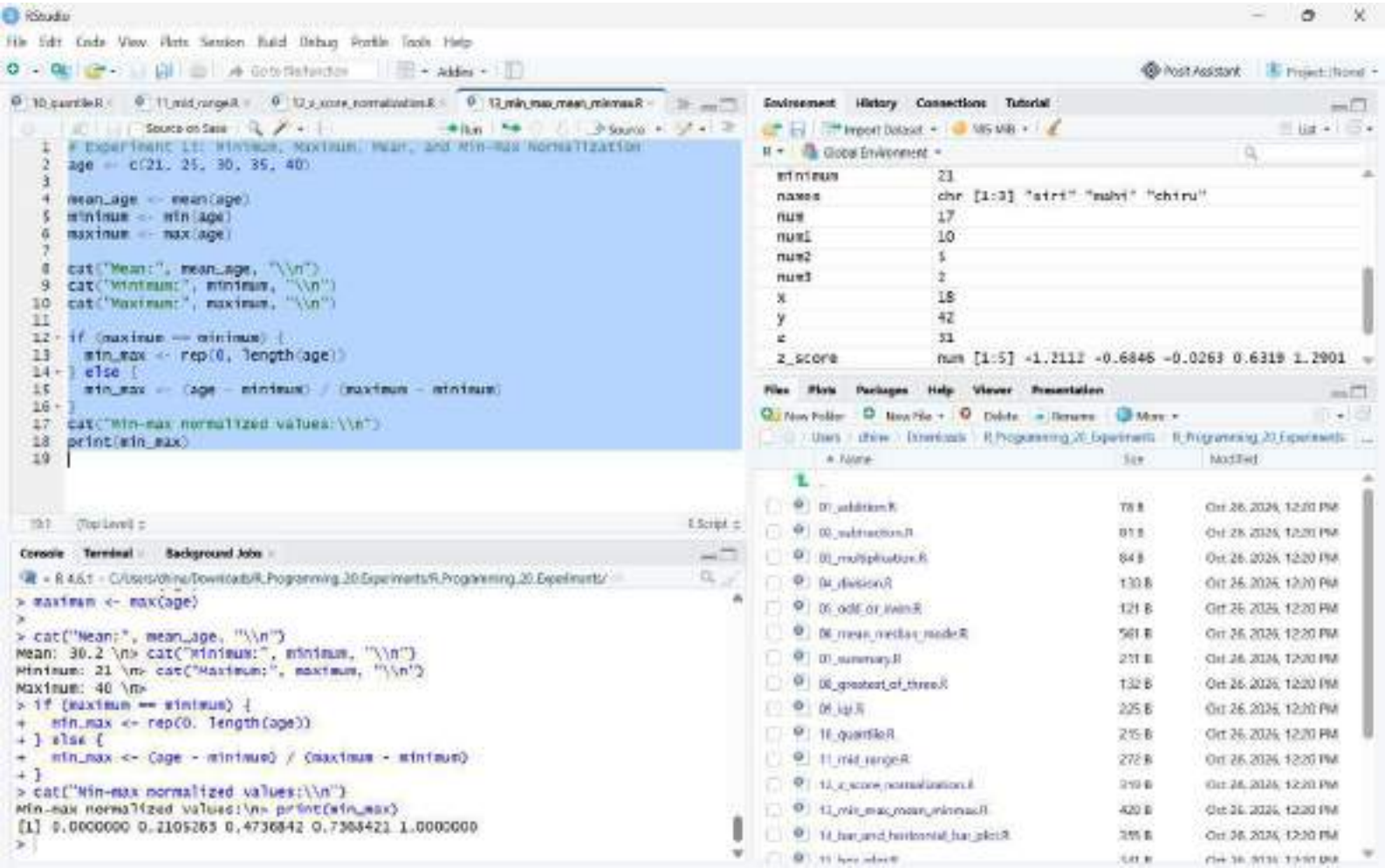


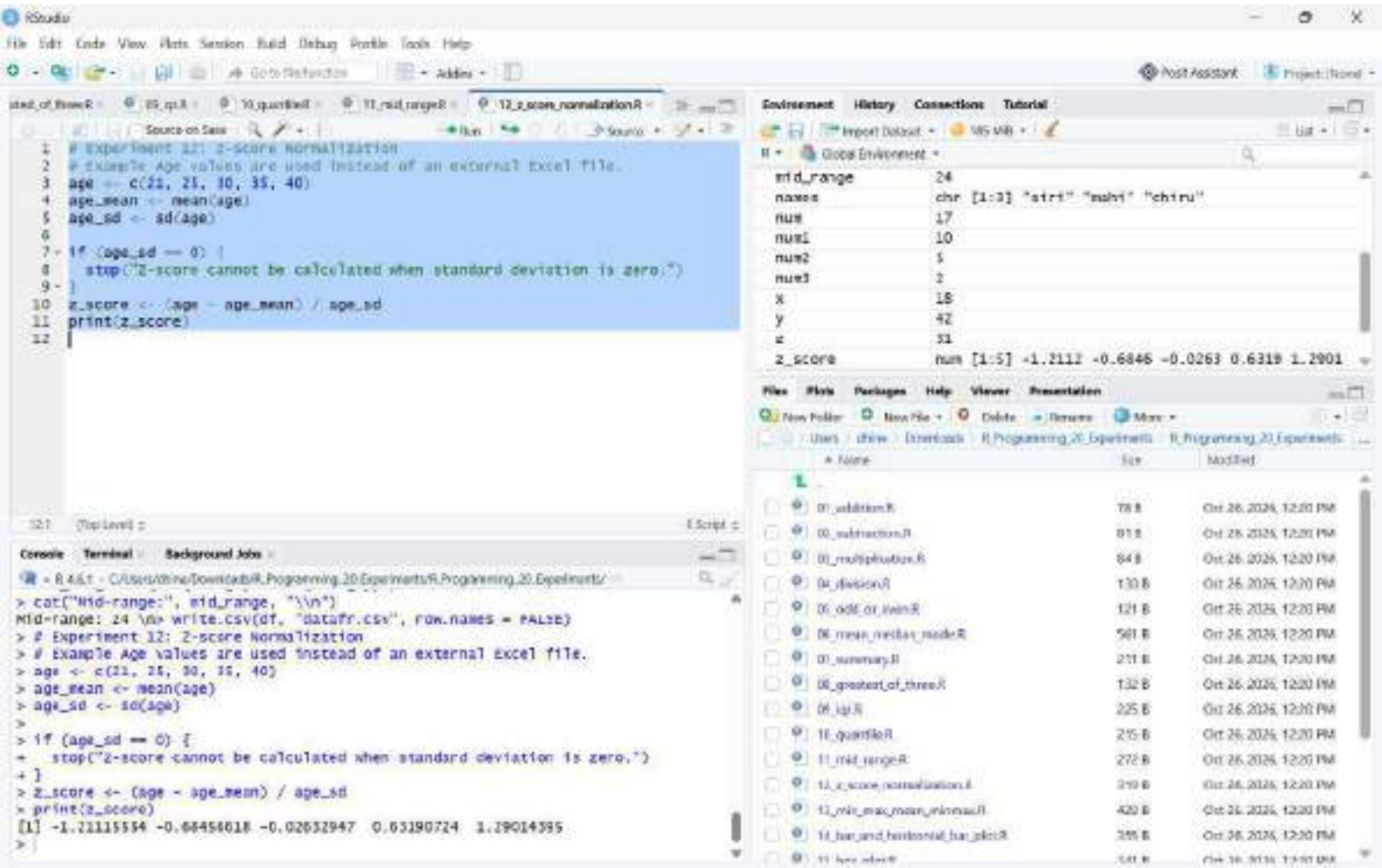


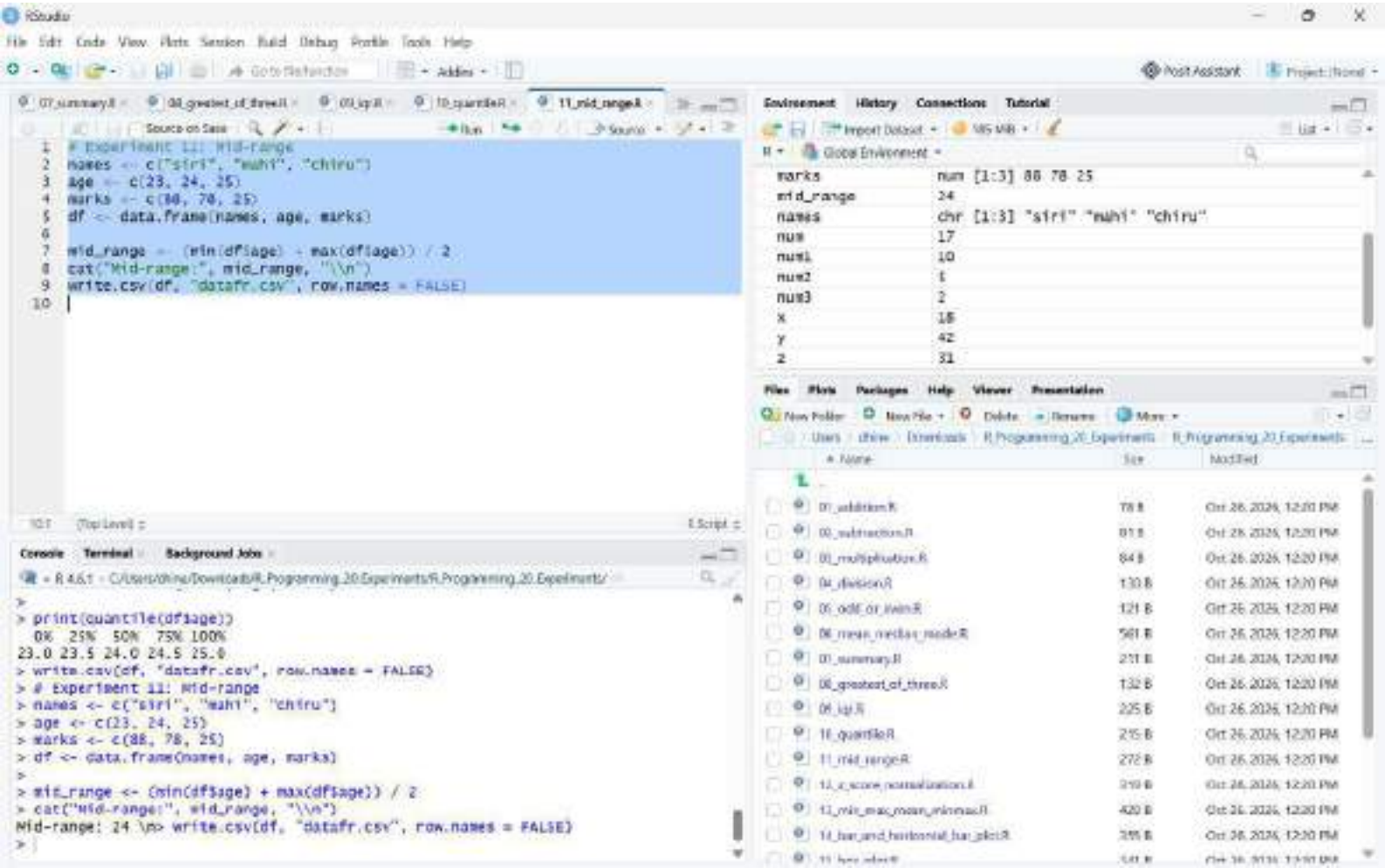
Console Terminal Background Jobs

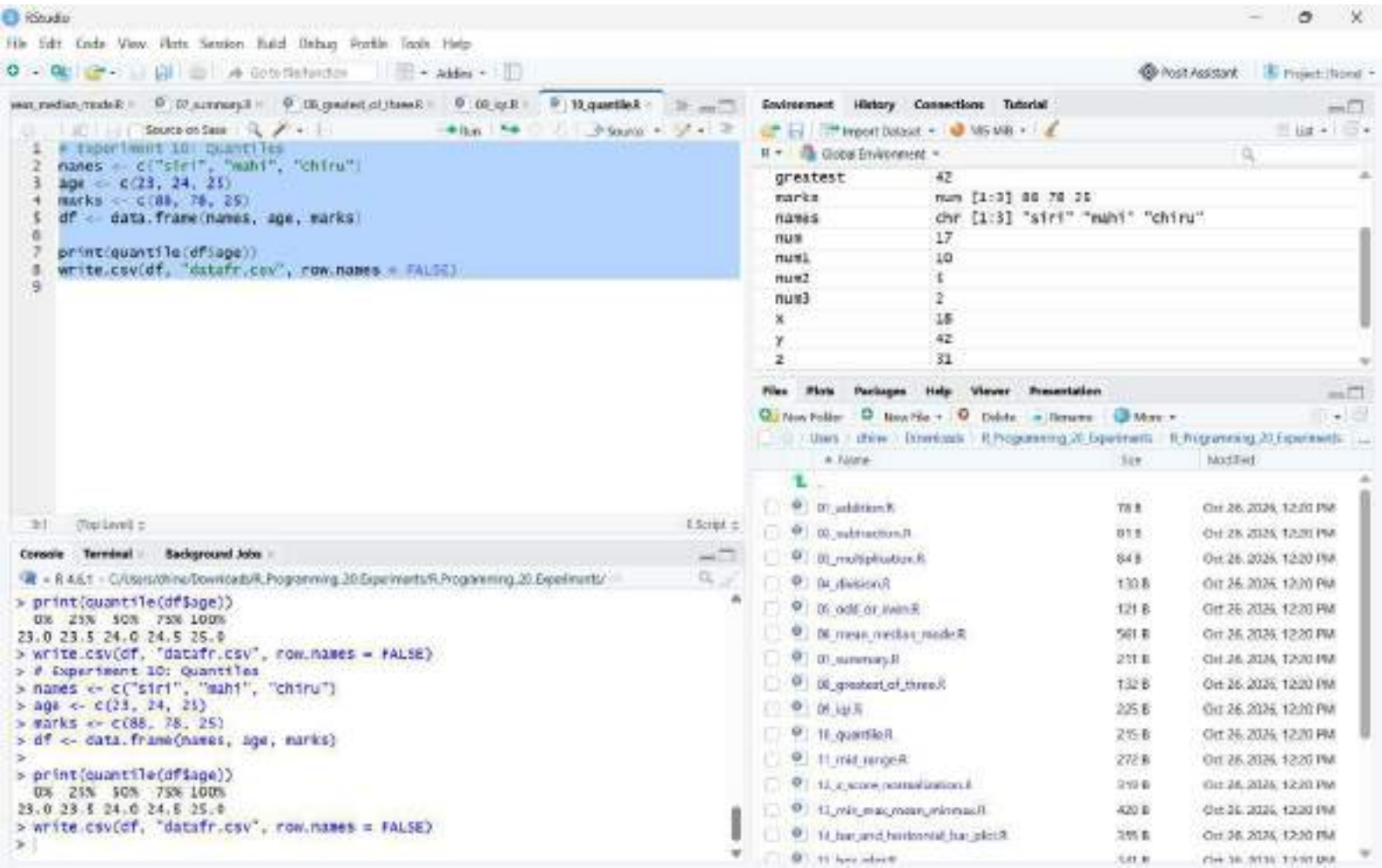
R 4.4.1 · C:/Users/dhina/Downloads/R Programming 20 Experiments/R Programming 20 Experiments/

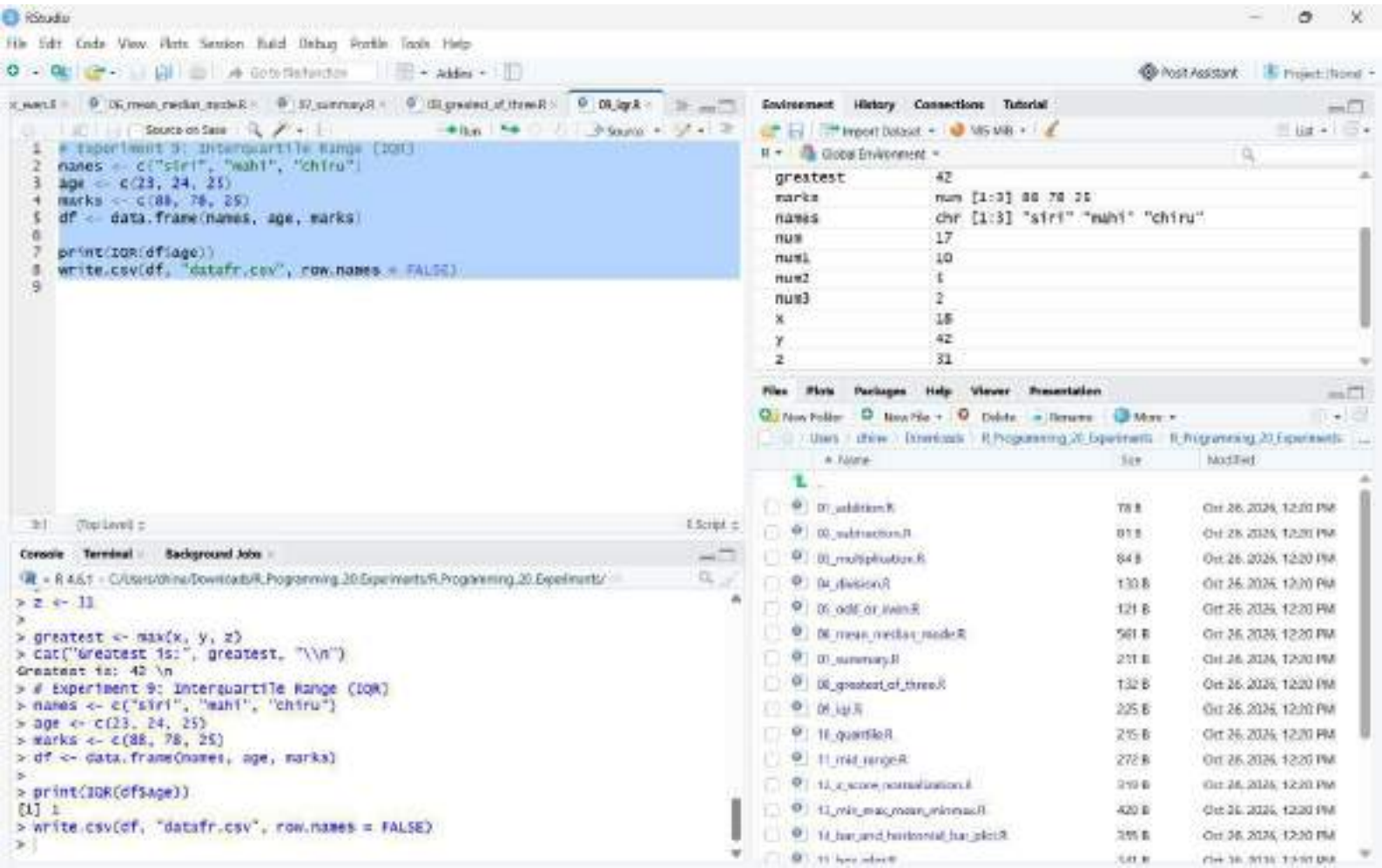
```
+     min_max <- (age - minimum) / (maximum - minimum)
+ }
> cat("Min-max normalized values:\\n")
Min-max normalized values:\n> print(min_max)
[1] 0.0000000 0.2105263 0.4736842 0.7368421 1.0000000
> # Experiment 14: Vertical and Horizontal Bar Plots
> a <- c(55, 67, 89, 80, 90)
>
> par(mfrow = c(1, 2))
> barplot(a, main = "Bar Plot", xlab = "Item", ylab = "Value",
+         names.arg = paste0("Item ", seq_along(a)))
> barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
+         xlab = "Value", names.arg = paste0("Item ", seq_along(a)))
> par(mfrow = c(1, 1))
>
```

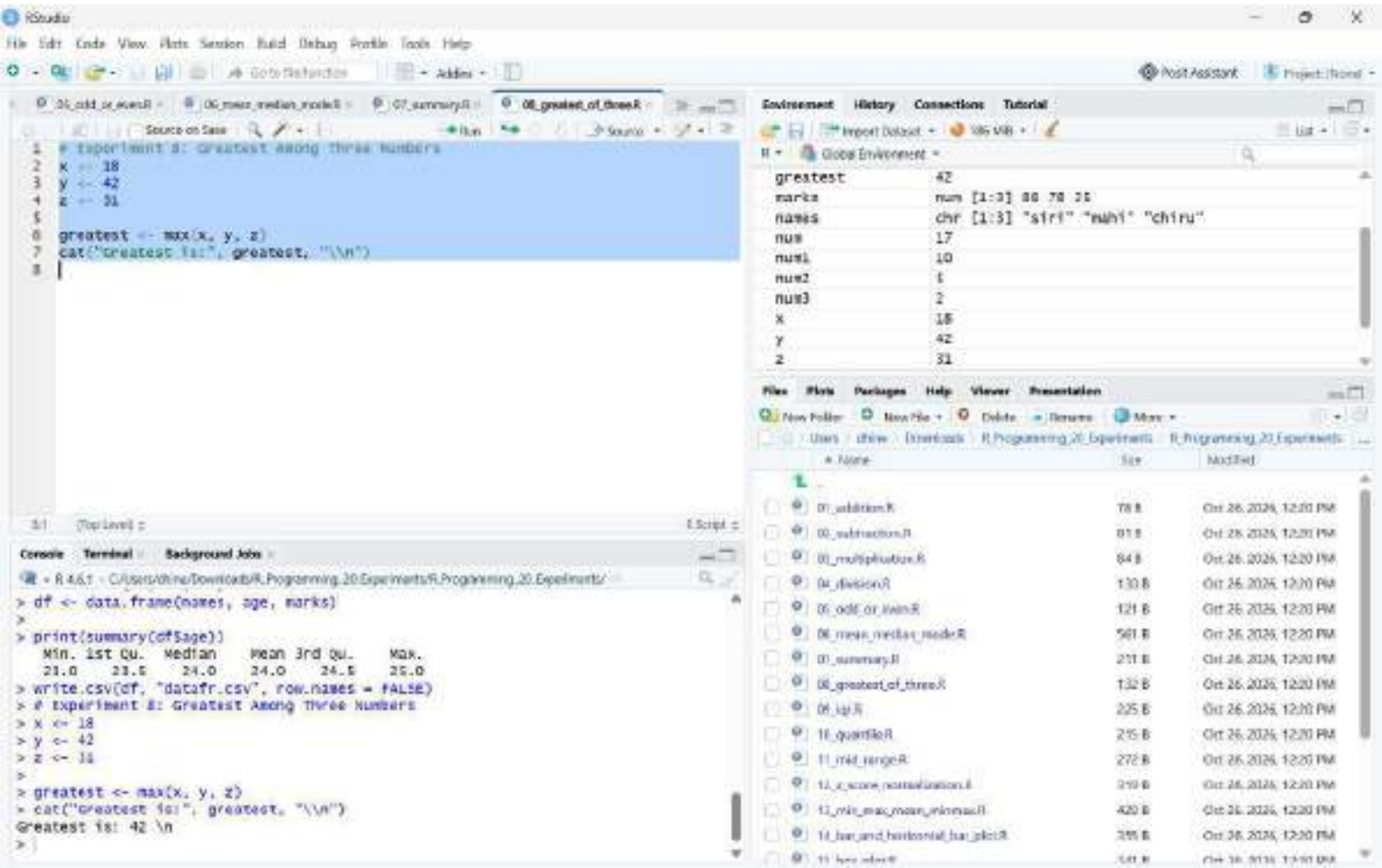

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
10_quantile.R
11_mid_range.R
12_z_score_normalization.R
13_min_max_mean_minmax.R
Experiment 13: Minimum, Maximum, Mean, and Min-Max Normalization
age <- c(21, 25, 30, 35, 40)
mean_age <- mean(age)
minimum <- min(age)
maximum <- max(age)
cat("Mean:", mean_age, "\n")
cat("Minimum:", minimum, "\n")
cat("Maximum:", maximum, "\n")
if (maximum == minimum) {
min_max <- rep(0, length(age))
} else {
min_max <- (age - minimum) / (maximum - minimum)
}
cat("Min-max normalized values:\n")
print(min_max)
Console Terminal Background Jobs
> maximum <- max(age)
> cat("Mean:", mean_age, "\n")
Mean: 30.2 \n> cat("Minimum:", minimum, "\n")
Minimum: 21 \n> cat("Maximum:", maximum, "\n")
Maximum: 40 \n>
> if (maximum == minimum) {
+ min_max <- rep(0, length(age))
+ } else {
+ min_max <- (age - minimum) / (maximum - minimum)
+ }
> cat("Min-max normalized values:\n")
Min-max normalized values:\n> print(min_max)
[1] 0.0000000 0.2105263 0.4736842 0.7368421 1.0000000
Environment History Connections Tutorial
minimum 21
names chr [1:3] "sirt" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
c 31
z_score num [1:5] -1.2112 -0.6846 -0.0263 0.6319 1.2901
Files Plots Packages Help Viewer Presentation
01_addition.R
02_subtraction.R
03_multiplication.R
04_division.R
05_odd_or_even.R
06_mean_median_mode.R
07_summary.R
08_greatest_of_three.R
09_sp.R
10_quantile.R
11_mid_range.R
12_z_score_normalization.R
13_min_max_mean_minmax.R
14_bar_and_horizontal_bar_plot.R

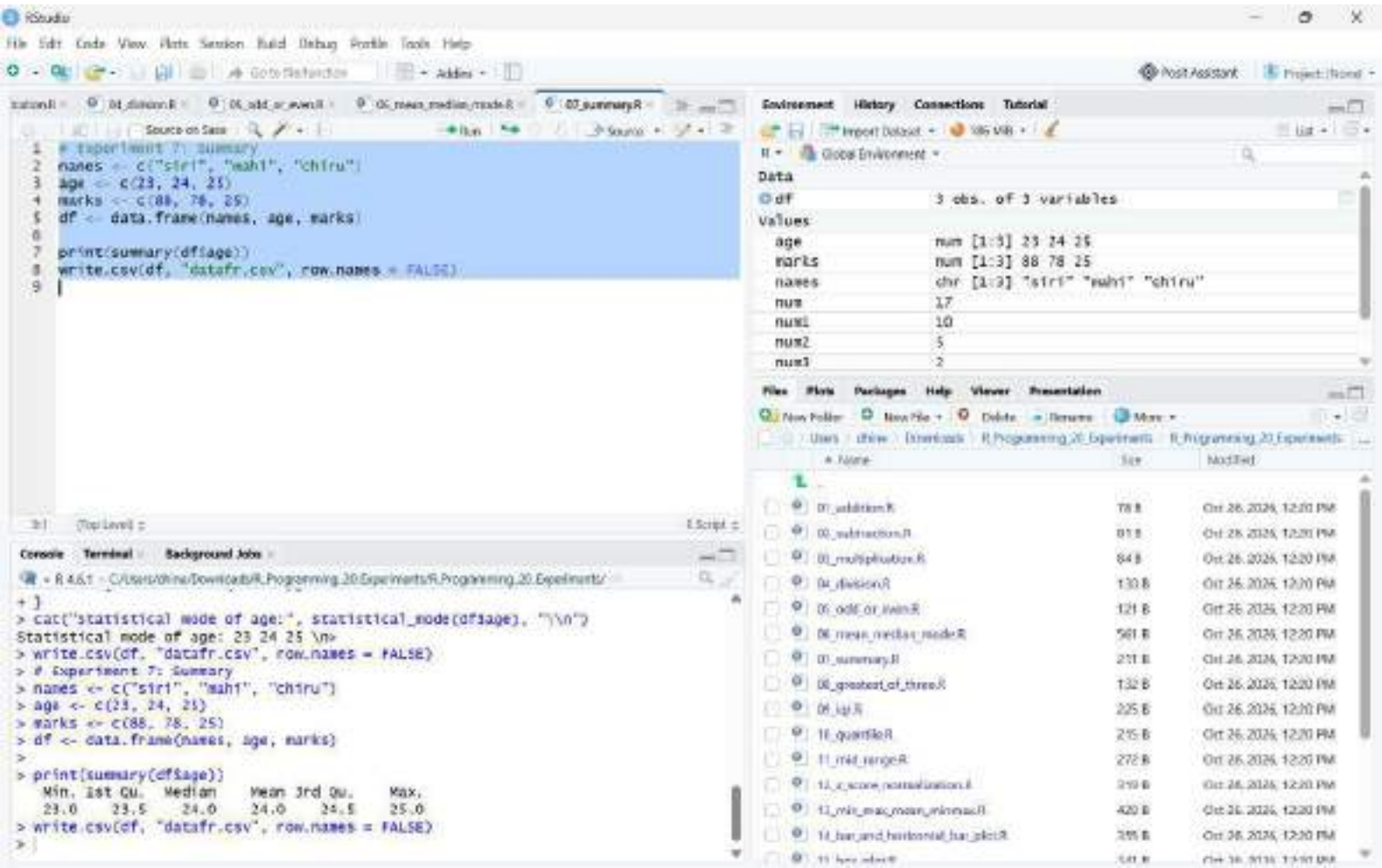

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Post Assistant
Project: None
12_z_score_normalization.R
Source on Save
Run
Source
Experiment 12: Z-score Normalization
Example Age values are used instead of an external Excel file.
age <- c(21, 25, 30, 35, 40)
age_mean <- mean(age)
age_sd <- sd(age)
if (age_sd == 0) {
stop("Z-score cannot be calculated when standard deviation is zero.")
}
z_score <- (age - age_mean) / age_sd
print(z_score)
Environment History Connections Tutorial
Import Dataset
Global Environment
mid_range 24
names chr [1:3] "sirt" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
z_score num [1:5] -1.2112 -0.6846 -0.0263 0.6319 1.2901
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Name
Size
Modified
Console Terminal Background Jobs
R 4.5.1 · C:/Users/dhine/Downloads/R Programming 20 Experiments/R Programming 20 Experiments/
> cat("Mid-range:", mid_range, "\\n")
Mid-range: 24 \n> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 12: Z-score Normalization
> # Example Age values are used instead of an external Excel file.
> age <- c(21, 25, 30, 35, 40)
> age_mean <- mean(age)
> age_sd <- sd(age)
>
> if (age_sd == 0) {
+ stop("Z-score cannot be calculated when standard deviation is zero.")
+ }
> z_score <- (age - age_mean) / age_sd
> print(z_score)
[1] -1.21115534 -0.68456618 -0.02632947 0.63190724 1.29014385
>

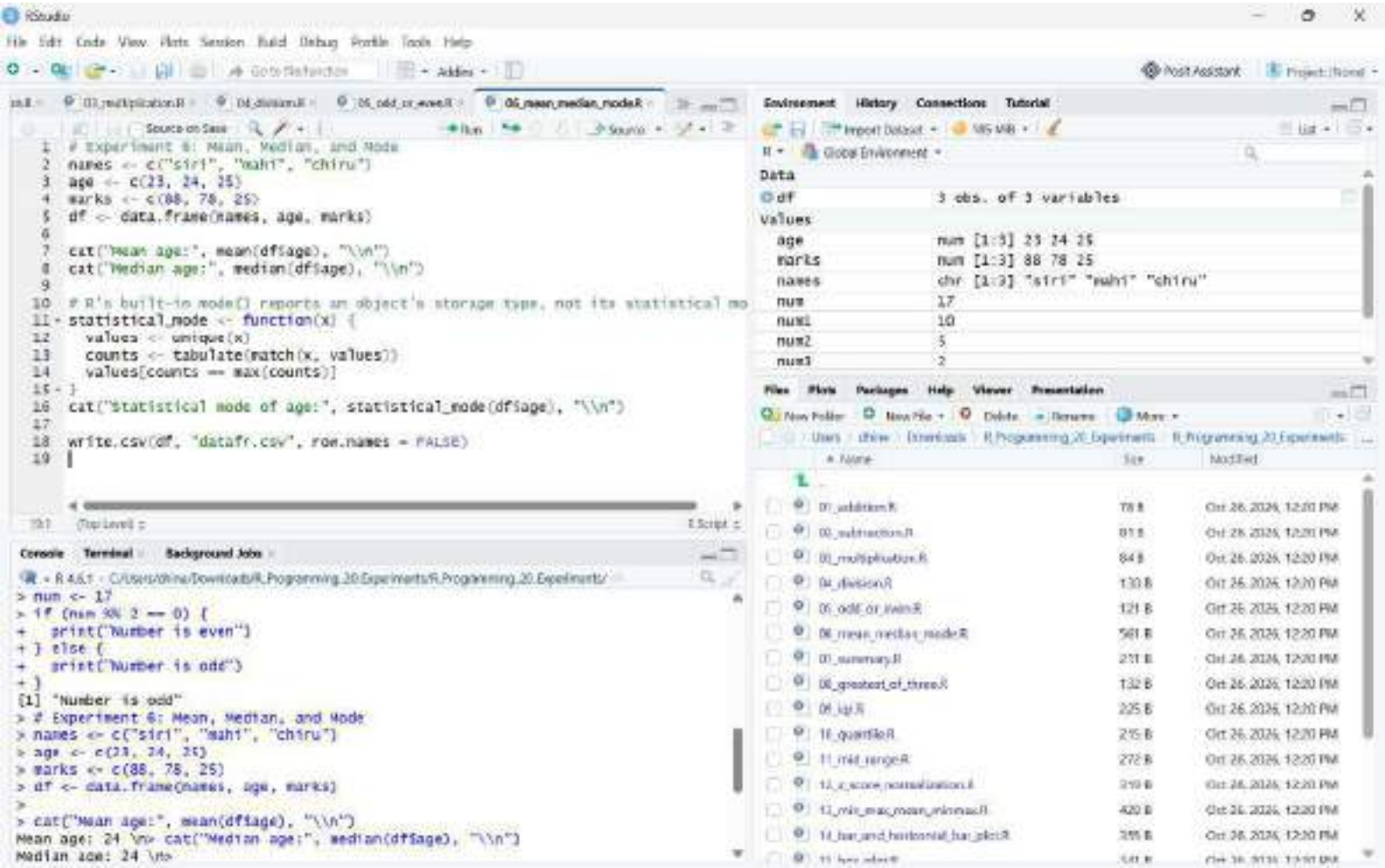

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
07_summary.R
08_greatest_of_three.R
09_iqr.R
10_quantile.R
11_mid_range.R
Source on Save
Run
Source
1 # Experiment 11: Mid-range
2 names <- c("siri", "mahi", "chiru")
3 age <- c(23, 24, 25)
4 marks <- c(88, 78, 25)
5 df <- data.frame(names, age, marks)
6
7 mid_range <- (min(df$age) + max(df$age)) / 2
8 cat("Mid-range:", mid_range, "\n")
9 write.csv(df, "datafr.csv", row.names = FALSE)
10
10:1 (Top Level)
R Script
Console Terminal Background Jobs
R 4.4.1 · C:/Users/dhina/Downloads/R Programming 20 Experiments/R Programming 20 Experiments/
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 11: Mid-range
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> mid_range <- (min(df$age) + max(df$age)) / 2
> cat("Mid-range:", mid_range, "\n")
Mid-range: 24 \n> write.csv(df, "datafr.csv", row.names = FALSE)
>
Environment History Connections Tutorial
Import Dataset
Global Environment
marks num [1:3] 88 78 25
mid_range 24
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 1
num3 2
x 18
y 42
z 31
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Name
Size
Modified
07_addition.R 78 B Oct 26, 2026, 12:20 PM
08_subtraction.R 81 B Oct 26, 2026, 12:20 PM
09_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 501 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 310 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 420 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 395 B Oct 26, 2026, 12:20 PM

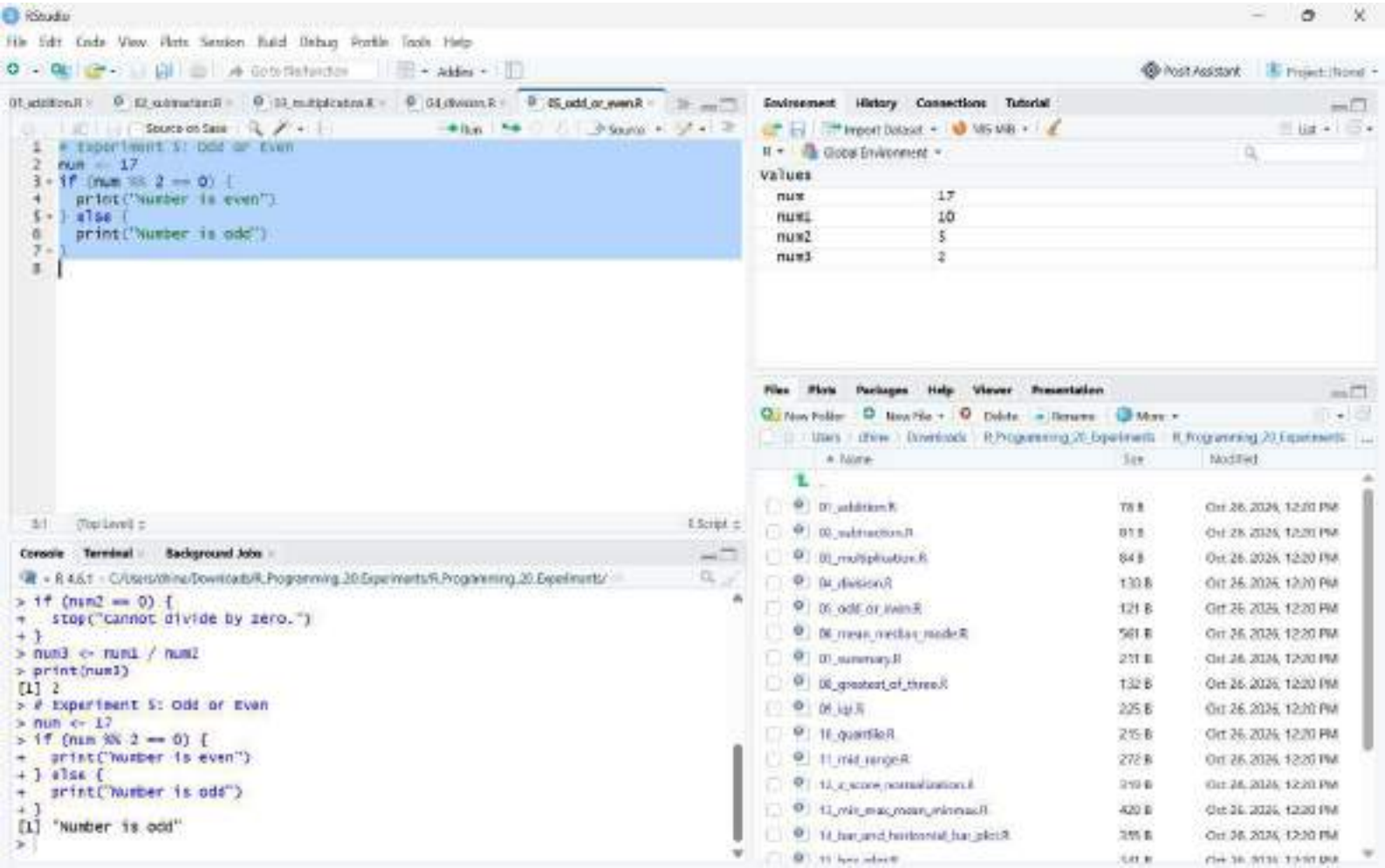

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
10_quantile.R
Experiment 10: Quantiles
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
print(quantile(df$age))
write.csv(df, "datafr.csv", row.names = FALSE)
Console Terminal Background Jobs
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 10: Quantiles
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
Environment History Connections Tutorial
greatest 42
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 1
num3 2
x 18
y 42
z 31
Files Plots Packages Help Viewer Presentation

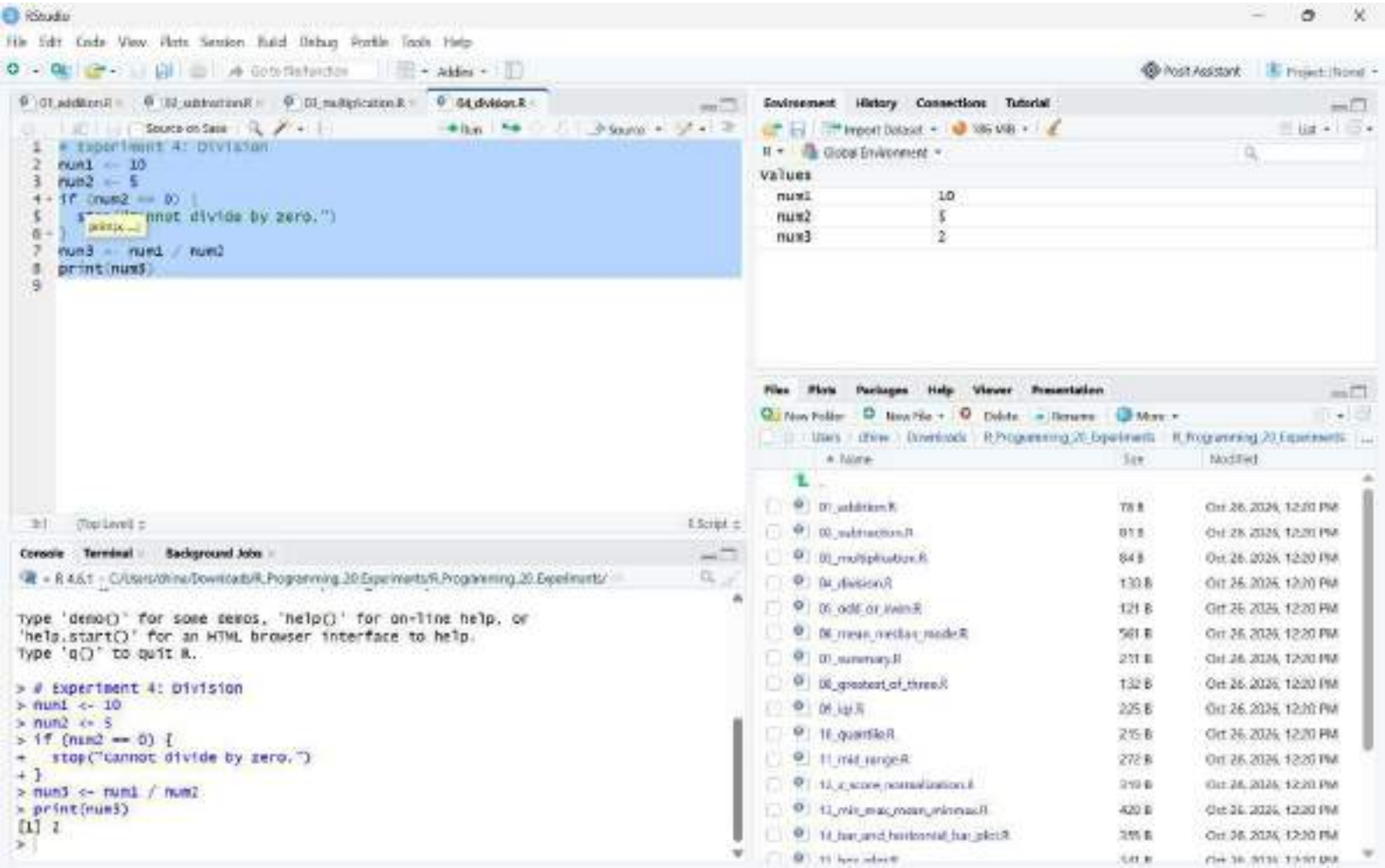

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Experiment 9: Interquartile Range (IQR)
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
print(IQR(df$age))
write.csv(df, "datafr.csv", row.names = FALSE)
> z <- 11
> greatest <- max(x, y, z)
> cat("Greatest is:", greatest, "\\n")
Greatest is: 42 \n
> # Experiment 9: Interquartile Range (IQR)
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> print(IQR(df$age))
[1] 1
> write.csv(df, "datafr.csv", row.names = FALSE)
greatest 42
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 1
num3 2
x 18
y 42
z 31

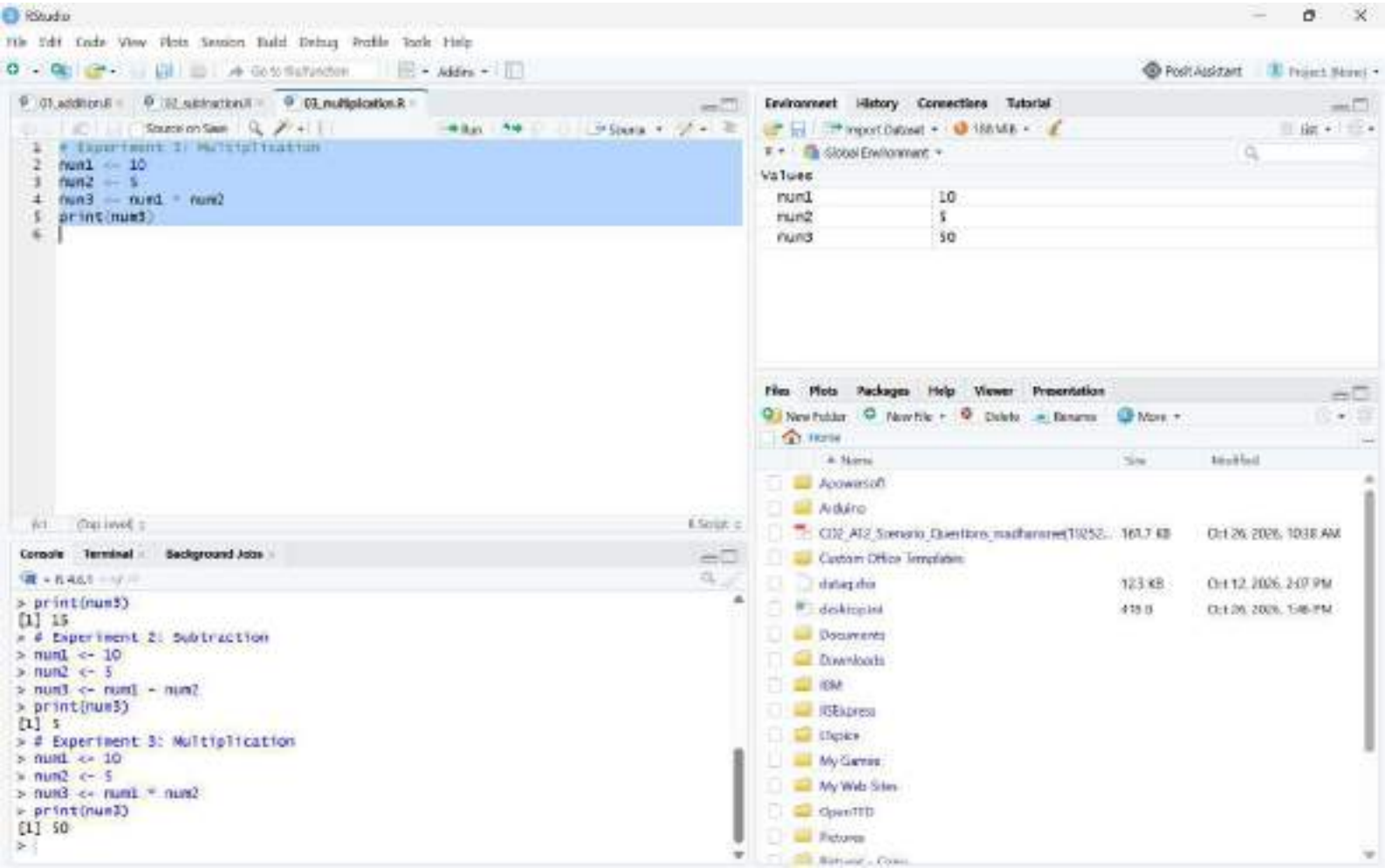

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
05_odd_or_even.R
06_mean_median_mode.R
07_summary.R
08_greatest_of_three.R
Source on Save
Run
Source
Experiment 8: Greatest Among Three Numbers
x <- 18
y <- 42
z <- 31
greatest <- max(x, y, z)
cat("Greatest is:", greatest, "\\n")
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.1 · C:/Users/dhine/Downloads/R Programming 20 Experiments/R Programming 20 Experiments/
> df <- data.frame(names, age, marks)
>
> print(summary(df$age))
Min. 1st Qu. Median Mean 3rd Qu. Max.
21.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 8: Greatest Among Three Numbers
> x <- 18
> y <- 42
> z <- 31
>
> greatest <- max(x, y, z)
> cat("Greatest is:", greatest, "\\n")
Greatest is: 42 \n
>
Environment History Connections Tutorial
Import Dataset
Global Environment
greatest 42
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 1
num3 2
x 18
y 42
z 31
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Name
Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 581 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_sp.R 225 B Oct 26, 2026, 12:20 PM
10_quartile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 310 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 420 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 395 B Oct 26, 2026, 12:20 PM

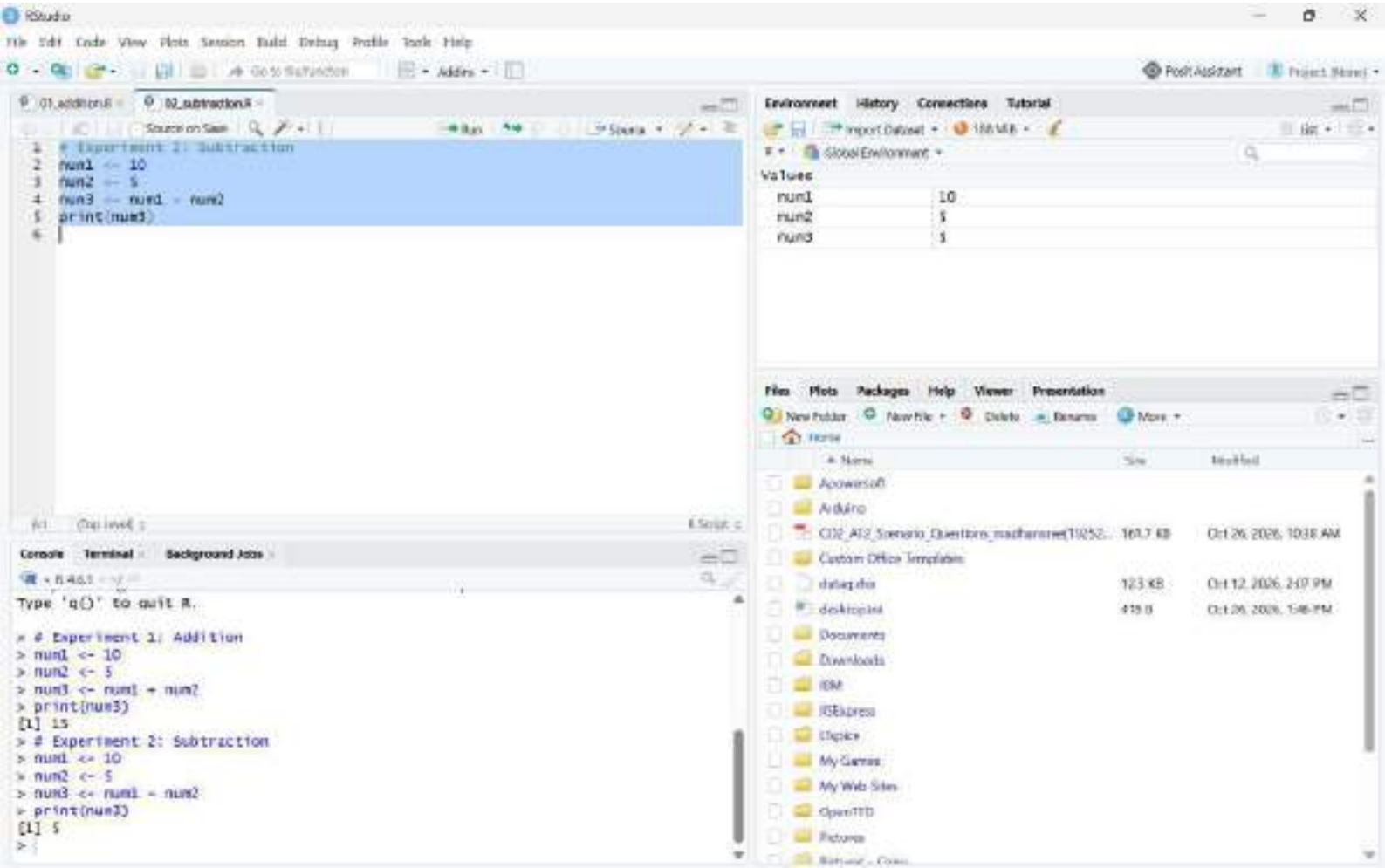

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
07_summary.R
Experiment 7: Summary
names <- c("sirf", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
print(summary(df$age))
write.csv(df, "datafr.csv", row.names = FALSE)
Environment History Connections Tutorial
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "sirf" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
> cat("statistical mode of age:", statistical_mode(df$age), "\n")
Statistical mode of age: 23 24 25 \n>
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 7: Summary
> names <- c("sirf", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> print(summary(df$age))
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)

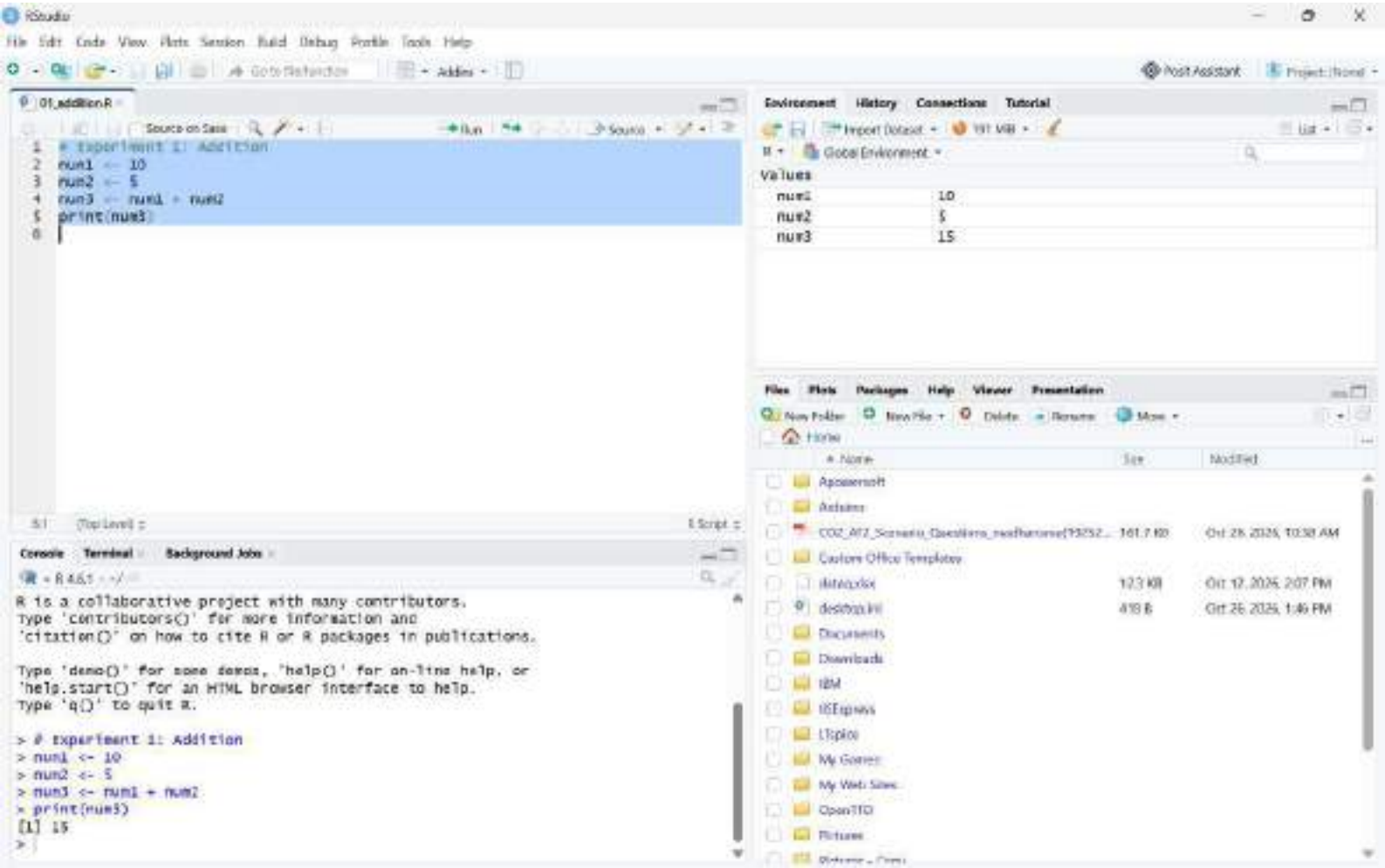

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
06_mean_median_mode.R
Experiment 6: Mean, Median, and Mode
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
cat("Mean age:", mean(df$age), "\\n")
cat("Median age:", median(df$age), "\\n")
R's built-in mode() reports an object's storage type, not its statistical mo
statistical_mode <- function(x) {
values <- unique(x)
counts <- tabulate(match(x, values))
values[counts == max(counts)]
}
cat("Statistical mode of age:", statistical_mode(df$age), "\\n")
write.csv(df, "datafr.csv", row.names = FALSE)
Environment History Connections Tutorial
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
> num <- 17
> if (num %% 2 == 0) {
+ print("Number is even")
+ } else {
+ print("Number is odd")
+ }
[1] "Number is odd"
> # Experiment 6: Mean, Median, and Mode
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> cat("Mean age:", mean(df$age), "\\n")
Mean age: 24 \n> cat("Median age:", median(df$age), "\\n")
Median age: 24 \n>


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
05_odd_or_even.R
Source on Save
Run
Source
Experiment 5: Odd or Even
num <- 17
if (num %% 2 == 0) {
print("Number is even")
} else {
print("Number is odd")
}
Environment History Connections Tutorial
Import Dataset
Global Environment
Values
num 17
num1 10
num2 5
num3 2
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Name
Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 581 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_sqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 420 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 395 B Oct 26, 2026, 12:20 PM
Console Terminal Background Jobs
R 4.6.1 · C:/Users/dhina/Downloads/R_Programming_20 Experiments/R_Programming_20 Experiments/
> if (num2 == 0) {
+ stop("Cannot divide by zero.")
+ }
> num3 <- num1 / num2
> print(num3)
[1] 2
> # Experiment 5: Odd or Even
> num <- 17
> if (num %% 2 == 0) {
+ print("Number is even")
+ } else {
+ print("Number is odd")
+ }
[1] "Number is odd"
>


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
04_division.R
Experiment 4: Division
num1 <- 10
num2 <- 5
if (num2 == 0) {
stop("Cannot divide by zero.")
}
num3 <- num1 / num2
print(num3)
Environment History Connections Tutorial
Values
num1 10
num2 5
num3 2
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> # Experiment 4: Division
> num1 <- 10
> num2 <- 5
> if (num2 == 0) {
+ stop("Cannot divide by zero.")
+ }
> num3 <- num1 / num2
> print(num3)
[1] 2


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
01_addition.R
02_subtraction.R
03_multiplication.R
Source on Save
Run
Source
1 # Experiment 3: Multiplication
2 num1 <- 10
3 num2 <- 5
4 num3 <- num1 * num2
5 print(num3)
6
Environment History Connections Tutorial
Import Dataset
Global Environment
Values
num1 10
num2 5
num3 50
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
Apowersoft
Arduino
Custom Office Templates
Documents
Downloads
IBM
IISExpress
My Games
My Web Sites
OpenTTD
Pictures
Console Terminal Background Jobs
> print(num3)
[1] 15
> # Experiment 2: Subtraction
> num1 <- 10
> num2 <- 5
> num3 <- num1 - num2
> print(num3)
[1] 5
> # Experiment 3: Multiplication
> num1 <- 10
> num2 <- 5
> num3 <- num1 * num2
> print(num3)
[1] 50
>


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
01_addition.R
02_subtraction.R
Source on Save
Run
Source
1 # Experiment 2: Subtraction
2 num1 <- 10
3 num2 <- 5
4 num3 <- num1 - num2
5 print(num3)
6
Environment History Connections Tutorial
Import Dataset
Global Environment
Values
num1 10
num2 5
num3 5
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
Apowersoft
Arduino
Custom Office Templates
Documents
Downloads
IBM
IISExpress
Oqtice
My Games
My Web Sites
OpenTTD
Pictures
Console Terminal Background Jobs
Type 'q()' to quit R.
> # Experiment 1: Addition
> num1 <- 10
> num2 <- 5
> num3 <- num1 + num2
> print(num3)
[1] 15
> # Experiment 2: Subtraction
> num1 <- 10
> num2 <- 5
> num3 <- num1 - num2
> print(num3)
[1] 5
>


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
01_addition.R
Source on Save
Run
Source
1 # Experiment 1: Addition
2 num1 <- 10
3 num2 <- 5
4 num3 <- num1 + num2
5 print(num3)
6
Environment History Connections Tutorial
Import Dataset
Global Environment
Values
num1 10
num2 5
num3 15
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
Apowersoft
Arduino
Custom Office Templates
desktop.ini 418 B Oct 26, 2025, 1:46 PM
Documents
Downloads
IBM
IISExpress
LTspice
My Games
My Web Sites
OpenTFO
Pictures
Console Terminal Background Jobs
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> # Experiment 1: Addition
> num1 <- 10
> num2 <- 5
> num3 <- num1 + num2
> print(num3)
[1] 15
>